The Coh Category: Projection-Induced Cost and Metric Enrichment

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Abstract

We define the Coh category, a verifier-certified categorical structure in which morphisms are generated by admissible traces rather than assumed a priori. The construction separates syntactic transition bookkeeping from semantic cost induced by projection from hidden realizations. Given a projection from hidden realizations to observable traces and an additive hidden cost functional, the observable envelope defect is defined as the supremum of hidden cost over the fiber of an observable trace. This envelope is subadditive under observable composition. Certified endpoint-typed morphisms whose declared defect bounds dominate the envelope compose into a category with locally preordered hom-sets. We then introduce a directed extended metric by taking the infimum of projection-induced defect over all admissible traces between two states. This yields a Lawvere-style metric enrichment of the Coh category. Finally, we define metric growth balls and a corresponding growth dimension, giving a precise route from verification-constrained trace composition to admissible geometry.

Keywords: verifier-certified traces; Lawvere metric; enriched categories; projection-induced cost; semantic defect; resource-bounded computation; admissible geometry.

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1 Introduction

In ordinary category-theoretic modeling, morphisms are typically specified first and then decorated with additional structure such as cost, weight, probability, or enrichment. In verification-sensitive computation and resource-bounded dynamics, this order is structurally backwards. A transition should not count as an available arrow merely because a source and target can be named. It should count as a morphism only after passing an admissibility test.

The Coh framework formalizes this reversal:

$$\text{proposal} \longrightarrow \text{verification} \longrightarrow \text{morphism}.$$

The central admissibility inequality is

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau), \tag{1}$$

where V is a valuation on states, Spend is cost paid by the transition, and Defect is admissible slack, tolerance, uncertainty, or projection loss. The two channels are not interchangeable.

The main contribution of this paper is a single formal spine:

$$\text{hidden realizations} \longrightarrow \text{projection} \longrightarrow \text{envelope defect} \longrightarrow \text{certified morphisms} \longrightarrow \text{metric enrichment}.$$

The paper proves that projection-induced semantic cost is subadditive under composition, that endpoint-typed certified morphisms form a category, and that the resulting category carries a directed extended metric defined by minimal admissible envelope defect.

Relation to existing structures. Lawvere metric spaces interpret generalized distances as enriched categories over the monoidal poset $([0,\infty], \geq, +, 0)$ [1]. Enriched category theory provides the surrounding formal language [2]. The Coh construction differs in its order of generation: the trace is not first a morphism with a cost annotation; it becomes a morphism only after verifier acceptance and the resource inequality (1) are satisfied. Projection-induced cost then supplies the metric structure.

Status of the results. The categorical and metric results are proved from the axioms stated below. The final growth-dimension discussion is a derived definition and interpretation, not a theorem identifying a universal fractal law. Where empirical or physical scaling is discussed, it is explicitly labeled as a model hypothesis.

2 Primitive Projection System

2.1 Hidden and observable substrates

Definition 2.1 (Hidden substrate)**.** *A hidden trajectory system consists of a class of hidden realizations $\mathcal{G}$ and a partially defined concatenation operation*

$$\cdot : \mathcal{G} \times \mathcal{G} \to \mathcal{G}.$$

When $\xi_2 \cdot \xi_1$ is defined, ξ_1 is executed first and ξ_2 second.

Definition 2.2 (Observable substrate)**.** *An observable trace system consists of a set $\mathcal{V}$ of observable traces, a partially defined concatenation operation*

$$\odot : \mathcal{V} \times \mathcal{V} \to \mathcal{V},$$

and a distinguished identity trace $e \in \mathcal{V}$.

Definition 2.3 (Projection and hidden cost)**.** *We fix a projection map*

$$\Pi : \mathcal{G} \to \mathcal{V}$$

and a nonnegative hidden cost functional

$$W : \mathcal{G} \to \mathbb{R}_{\geq 0}.$$

2.2 Structural axioms

Axiom 2.1 (Observable associativity)**.** *For all composable $R_1, R_2, R_3 \in \mathcal{V}$,*

$$(R_3 \odot R_2) \odot R_1 = R_3 \odot (R_2 \odot R_1).$$

Axiom 2.2 (Observable identity)**.** *For every $R \in \mathcal{V}$,*

$$e \odot R = R, \qquad R \odot e = R$$

whenever either side is defined.

Axiom 2.3 (Projection compatibility)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$\Pi(\xi_2 \cdot \xi_1) = \Pi(\xi_2) \odot \Pi(\xi_1).$$

Axiom 2.4 (Fiber decomposition)**.** *If $\xi \in \mathcal{G}$ satisfies*

$$\Pi(\xi) = R_2 \odot R_1,$$

then there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Axiom 2.5 (Additivity of hidden cost)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$W(\xi_2 \cdot \xi_1) = W(\xi_1) + W(\xi_2).$$

Axiom 2.6 (Identity fiber nonempty and zero-cost)**.**

$$\Pi^{-1}(e) \neq \emptyset, \qquad \forall \xi \in \Pi^{-1}(e),\ W(\xi) = 0.$$

3 Envelope Defect

Definition 3.1 (Fiber)**.** *For $R \in \mathcal{V}$, define the hidden fiber*

$$\mathcal{F}_R := \{\xi \in \mathcal{G} : \Pi(\xi) = R\}.$$

Definition 3.2 (Admissible observable trace)**.** *A trace $R \in \mathcal{V}$ is admissible if $\mathcal{F}_R \neq \emptyset$. Write $\mathcal{V}_{\mathrm{adm}} \subseteq \mathcal{V}$ for the set of admissible observable traces.*

Axiom 3.1 (Fiber-boundedness)**.** *For every $R \in \mathcal{V}_{\mathrm{adm}}$,*

$$\sup_{\xi \in \mathcal{F}_R} W(\xi) < \infty.$$

Definition 3.3 (Envelope defect)**.** *For each $R \in \mathcal{V}_{\mathrm{adm}}$, define*

$$\delta(R) := \sup_{\xi \in \mathcal{F}_R} W(\xi). \tag{2}$$

Remark 3.1. *All suprema in this paper are taken in $\mathbb{R}_{\geq 0}$ after restricting to admissible traces. Axiom 3.1 ensures that the envelope defect is finite on $\mathcal{V}_{\mathrm{adm}}$.*

Lemma 3.1 (Envelope soundness)**.** *For every $\xi \in \mathcal{G}$ such that $\Pi(\xi) \in \mathcal{V}_{\mathrm{adm}}$,*

$$W(\xi) \leq \delta(\Pi(\xi)).$$

Proof. By definition of fiber, $\xi \in \mathcal{F}_{\Pi(\xi)}$. Since $\delta(\Pi(\xi))$ is the supremum of W over this fiber, it is an upper bound for $W(\xi)$. □

Lemma 3.2 (Envelope minimality)**.** *Let $\Delta : \mathcal{V}_{\mathrm{adm}} \to \mathbb{R}_{\geq 0}$ satisfy*

$$W(\xi) \leq \Delta(\Pi(\xi))$$

for all hidden realizations ξ whose projection is admissible. Then for every $R \in \mathcal{V}_{\mathrm{adm}}$,

$$\delta(R) \leq \Delta(R).$$

Proof. Fix $R \in \mathcal{V}_{\mathrm{adm}}$. For each $\xi \in \mathcal{F}_R$, one has $\Pi(\xi) = R$, and therefore $W(\xi) \leq \Delta(R)$. Thus $\Delta(R)$ is an upper bound for the set $\{W(\xi) : \xi \in \mathcal{F}_R\}$. By definition of supremum, $\delta(R) \leq \Delta(R)$. □

Lemma 3.3 (Identity envelope)**.**

$$\delta(e) = 0.$$

Proof. By Axiom 2.6, the identity fiber $\mathcal{F}_e = \Pi^{-1}(e)$ is nonempty and every element in that fiber has hidden cost zero. Hence the set $\{W(\xi) : \xi \in \mathcal{F}_e\}$ is exactly $\{0\}$. Its supremum is zero. □

Lemma 3.4 (Projection-induced subadditivity)**.** *Let $R_1, R_2 \in \mathcal{V}_{\mathrm{adm}}$, and assume that $R_2 \odot R_1$ is admissible. Then*

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2). \tag{3}$$

Proof. Let $\xi \in \mathcal{F}_{R_2 \odot R_1}$. By Axiom 2.4, there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Thus $\xi_i \in \mathcal{F}_{R_i}$. By Axiom 2.5,

$$W(\xi) = W(\xi_2 \cdot \xi_1) = W(\xi_1) + W(\xi_2).$$

By Lemma 3.1, $W(\xi_1) \leq \delta(R_1)$ and $W(\xi_2) \leq \delta(R_2)$. Hence

$$W(\xi) \leq \delta(R_1) + \delta(R_2).$$

The right-hand side is therefore an upper bound for W over $\mathcal{F}_{R_2 \odot R_1}$. Taking the supremum over that fiber gives (3). □

4 Endpoint-Typed Certified Morphisms

Definition 4.1 (Coh boundary)**.** *A Coh boundary object X consists of a carrier set Σ_X and a valuation*

$$V_X : \Sigma_X \to \mathbb{R}_{\geq 0}.$$

Definition 4.2 (Endpoint compatibility)**.** *A trace $\tau \in \mathcal{V}_{\mathrm{adm}}$ is compatible with endpoints $x \in \Sigma_X$ and $y \in \Sigma_Y$ if it is assigned a typing relation*

$$\tau : x \rightsquigarrow y.$$

The relation $\rightsquigarrow$ is treated as primitive in this foundation layer.

Definition 4.3 (Certified morphism)**.** *Let X, Y be Coh boundary objects. A certified morphism $f : X \to Y$ is a tuple*

$$f = (\tau_f, s_f, d_f, x, y)$$

such that

(i) $x \in \Sigma_X$ *and* $y \in \Sigma_Y$*;*

(ii) $\tau_f \in \mathcal{V}_{\mathrm{adm}}$*;*

(iii) $\tau_f : x \rightsquigarrow y$*;*

(iv) $d_f \geq \delta(\tau_f)$*;*

(v) $V_Y(y) + s_f \leq V_X(x) + d_f$*.*

Here s_f is certified spend and d_f is certified defect.

Definition 4.4 (Composition of certified morphisms)**.** *Let*

$$f = (\tau_f, s_f, d_f, x, y) : X \to Y, \qquad g = (\tau_g, s_g, d_g, y, z) : Y \to Z.$$

Assume $\tau_g \odot \tau_f$ is defined, admissible, and typed

$$\tau_g \odot \tau_f : x \rightsquigarrow z.$$

Define

$$g \circ f := (\tau_g \odot \tau_f,\ s_f + s_g,\ d_f + d_g,\ x,\ z).$$

Definition 4.5 (Identity morphism)**.** *For each Coh boundary object X and each $x \in \Sigma_X$, define*

$$\mathrm{id}_x := (e, 0, 0, x, x).$$

Remark 4.1 (State-indexing)**.** *Because identities are indexed by states $x \in \Sigma_X$, one may equivalently regard the base objects of the category as state-labeled boundaries (X, x). We keep the boundary notation because it exposes the valuation map V_X.*

5 Certified Category Theorem

Theorem 5.1 (Closure under certified composition)**.** *Let*

$$f = (\tau_f, s_f, d_f, x, y) : X \to Y, \qquad g = (\tau_g, s_g, d_g, y, z) : Y \to Z$$

be certified morphisms. Assume $\tau_g \odot \tau_f$ is defined, admissible, and typed $\tau_g \odot \tau_f : x \rightsquigarrow z$. Then $g \circ f$ is certified.

Proof. First, by Lemma 3.4,

$$\delta(\tau_g \odot \tau_f) \le \delta(\tau_f) + \delta(\tau_g).$$

Since f and g are certified,

$$d_f \ge \delta(\tau_f), \qquad d_g \ge \delta(\tau_g).$$

Therefore

$$d_f + d_g \ge \delta(\tau_f) + \delta(\tau_g) \ge \delta(\tau_g \odot \tau_f),$$

so the defect bound for the composite holds.

Next, the certified inequalities for f and g are

$$V_Y(y) + s_f \le V_X(x) + d_f, \qquad V_Z(z) + s_g \le V_Y(y) + d_g.$$

Adding gives

$$V_Y(y) + V_Z(z) + s_f + s_g \le V_X(x) + V_Y(y) + d_f + d_g.$$

Since all valuations are finite nonnegative real numbers, cancel $V_Y(y)$ to obtain

$$V_Z(z) + (s_f + s_g) \le V_X(x) + (d_f + d_g).$$

This is the valuation inequality required for $g \circ f$. The endpoint, admissibility, and typing requirements hold by the hypotheses of Definition 4.4. □

Theorem 5.2 (Certified identity)**.** *For every Coh boundary object X and every $x \in \Sigma_X$, $\mathrm{id}_x = (e, 0, 0, x, x)$ is certified.*

Proof. By Axiom 2.6, e is admissible. By Lemma 3.3, $\delta(e) = 0$, so the defect condition is $0 \ge 0$. The valuation inequality is

$$V_X(x) + 0 \le V_X(x) + 0,$$

which is immediate. □

Theorem 5.3 (Associativity of certified composition)**.** *Let*

$$f = (\tau_f, s_f, d_f, w, x), \quad g = (\tau_g, s_g, d_g, x, y), \quad h = (\tau_h, s_h, d_h, y, z)$$

be composable certified morphisms, and assume all relevant composites are defined, admissible, and correctly typed. Then

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. By definition of certified composition,

$$(h \circ g) \circ f = ((\tau_h \odot \tau_g) \odot \tau_f,\ (s_f + s_g) + s_h,\ (d_f + d_g) + d_h,\ w,\ z),$$

whereas

$$h \circ (g \circ f) = (\tau_h \odot (\tau_g \odot \tau_f),\ s_f + (s_g + s_h),\ d_f + (d_g + d_h),\ w,\ z).$$

Axiom 2.1 identifies the trace components. Associativity of addition identifies the spend and defect components. Therefore the tuples are equal. □

Theorem 5.4 (Coh category)**.** *Certified boundary states and certified morphisms form a category, denoted* $\mathbf{Coh}_\delta$*.*

Proof. Closure of composition is Theorem 5.1. Certified identities are provided by Theorem 5.2. Associativity is Theorem 5.3. The identity laws follow from Axiom 2.2 and the fact that adding zero to spend and defect leaves the tuple unchanged. Hence the certified system forms a category. □

6 Local Preorder by Defect

Definition 6.1 (Hom-set preorder)**.** *For fixed Coh boundary objects* X, Y*, define*

$$f \preceq g \quad \Longleftrightarrow \quad d_f \leq d_g$$

for certified morphisms $f, g \in_{\mathbf{Coh}_\delta} (X, Y)$*.*

Theorem 6.1 (Hom-set preorder)**.** *For each fixed pair* X, Y*, the relation* $\preceq$ *is a preorder on* $_{\mathbf{Coh}_\delta}(X, Y)$*.*

Proof. Reflexivity follows from $d_f \leq d_f$. Transitivity follows because $d_f \leq d_g$ and $d_g \leq d_h$ imply $d_f \leq d_h$. □

Theorem 6.2 (Monotonicity of composition)**.** *If*

$$f_1 \preceq f_2, \qquad g_1 \preceq g_2,$$

and both composites are defined, then

$$g_1 \circ f_1 \preceq g_2 \circ f_2.$$

Proof. The assumptions mean $d_{f_1} \leq d_{f_2}$ and $d_{g_1} \leq d_{g_2}$. Adding gives

$$d_{f_1} + d_{g_1} \leq d_{f_2} + d_{g_2}.$$

By Definition 4.4, the defect component of a composite is the sum of the defect components. Therefore

$$d_{g_1 \circ f_1} \leq d_{g_2 \circ f_2},$$

which is precisely $g_1 \circ f_1 \preceq g_2 \circ f_2$. □

7 Metric Enrichment

We now pass from individual certified morphisms to a minimal admissible semantic distance. The construction is directed: in general $d(x, y)$ need not equal $d(y, x)$.

Definition 7.1 (Coh distance)**.** *For boundary states* $x \in \Sigma_X$ *and* $y \in \Sigma_Y$*, define*

$$d_{\mathrm{Coh}}(x, y) := \inf\{\delta(\tau) : \tau \in \mathcal{V}_{\mathrm{adm}},\ \tau : x \rightsquigarrow y\}. \tag{4}$$

If there is no admissible trace from x *to* y*, set* $d_{\mathrm{Coh}}(x, y) = +\infty$*.*

Remark 7.1. *The codomain of* d_{Coh} *is* $\mathbb{R}_{\geq 0} \cup \{+\infty\}$*. This is the natural codomain for Lawvere-style directed distances.*

Lemma 7.1 (Zero self-distance)**.** *For every boundary state x,*

$$d_{\mathrm{Coh}}(x, x) = 0.$$

Proof. The identity trace $e : x \rightsquigarrow x$ is admissible by Axiom 2.6. Lemma 3.3 gives $\delta(e) = 0$. Therefore the infimum in (4) is bounded above by zero. Since δ is nonnegative, the infimum is also bounded below by zero. Hence $d_{\mathrm{Coh}}(x, x) = 0$. □

Lemma 7.2 (Directed triangle inequality)**.** *For boundary states x, y, z,*

$$d_{\mathrm{Coh}}(x, z) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z).$$

Proof. If either $d_{\mathrm{Coh}}(x, y) = +\infty$ or $d_{\mathrm{Coh}}(y, z) = +\infty$, the right-hand side is $+\infty$ and the claim is immediate. Otherwise, fix $\varepsilon > 0$. By definition of infimum, choose admissible traces $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ such that

$$\delta(\tau_1) \leq d_{\mathrm{Coh}}(x, y) + \varepsilon/2, \qquad \delta(\tau_2) \leq d_{\mathrm{Coh}}(y, z) + \varepsilon/2.$$

By certified closure, the composite $\tau_2 \odot \tau_1 : x \rightsquigarrow z$ is admissible whenever the composition hypotheses are satisfied. By Lemma 3.4,

$$\delta(\tau_2 \odot \tau_1) \leq \delta(\tau_1) + \delta(\tau_2) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z) + \varepsilon.$$

Since $d_{\mathrm{Coh}}(x, z)$ is the infimum over admissible traces from x to z,

$$d_{\mathrm{Coh}}(x, z) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z) + \varepsilon.$$

Letting $\varepsilon \to 0$ gives the result. □

Theorem 7.1 (Metric enrichment)**.** *The state-indexed Coh category carries a directed extended pseudometric d_{Coh}. Equivalently, it is enriched in the Lawvere quantale $([0, \infty], \geq, +, 0)$ after interpreting composition through the triangle inequality.*

Proof. Lemma 7.1 gives the identity law $d_{\mathrm{Coh}}(x, x) = 0$. Lemma 7.2 gives the directed triangle inequality. Nonnegativity follows from nonnegativity of W and hence of δ. Symmetry is not assumed and generally fails. Thus d_{Coh} is a directed extended pseudometric. This is precisely the metric data of a Lawvere-enriched category. □

8 Metric Growth and Admissible Geometry

Definition 8.1 (Metric ball)**.** *For a state x and radius $r \geq 0$, define*

$$B_r(x) := \{y : d_{\mathrm{Coh}}(x, y) \leq r\}.$$

Definition 8.2 (Metric growth dimension)**.** *When the relevant state set is countable or discretized at a fixed observational resolution, define*

$$D_{\mathrm{Coh}}(x) := \limsup_{r \to \infty} \frac{\log |B_r(x)|}{\log r}.$$

Remark 8.1 (Interpretation)**.** *$D_{\mathrm{Coh}}(x)$ measures the growth rate of reachable states under bounded verified semantic cost. It is not asserted to be finite in full generality; finiteness depends on the state space, resolution, verifier, and cost envelope.*

8.1 Entropy-filtered scaling model

For branching systems, one may introduce an acceptance ratio

$$a := \frac{\#\{\text{accepted candidate continuations}\}}{\#\{\text{proposed candidate continuations}\}},$$

computed at a fixed scale or averaged over a growth window. A useful model hypothesis is that admissibility reduces branching entropy rather than merely changing a field exponent. This leads to the scaling ansatz

$$D_{\text{model}}(\eta, a) = 1 + \frac{a^\gamma}{1 + \beta\eta}, \qquad \beta, \gamma > 0, \tag{5}$$

where η is a growth-bias exponent and a is the Coh acceptance ratio.

Remark 8.2 (Status of the scaling law)**.** *Equation* (5) *is not used in the category or metric proofs. It is a calibration model for experimental systems such as electrical branching, dielectric breakdown, or verifier-filtered growth simulations. Its role is to connect metric growth to measurable branching geometry.*

9 Figures

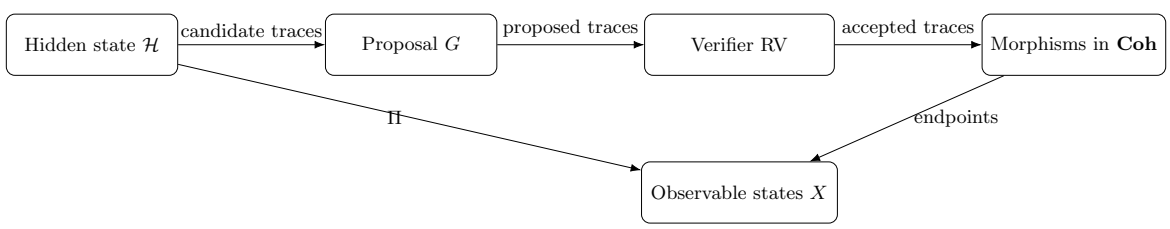

Figure 1: Coh execution wedge. Hidden-state proposals generate candidate traces, the verifier certifies a subset, and only certified traces enter the categorical layer.

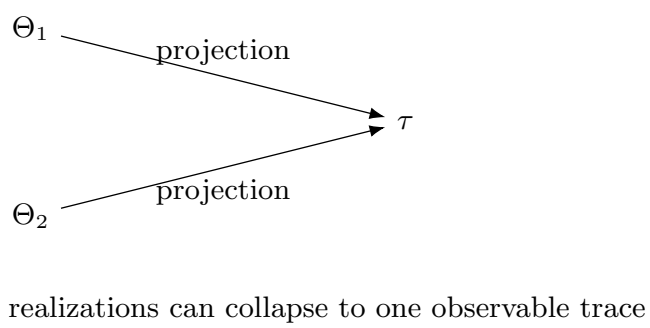

Figure 2: Projection from hidden realizations to observable traces. Multiple hidden realizations may collapse to the same observable trace, motivating the fiber envelope definition of δ.

10 Failure Modes and Boundary Cases

Unbounded fibers. If Axiom 3.1 fails, $\delta(R)$ may be infinite or undefined as a finite real value. The metric may remain extended-valued, but finite certification requires an explicit bound.

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

Figure 3: Certified composition. If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ are admissible and compatible, their composite is trace concatenation $\tau_2 \circ \tau_1 = \tau_2 \odot \tau_1$.

Projection failure. If the observable trace does not retain the information needed to construct a conservative fiber envelope, the verifier may certify a surrogate weaker than the hidden system. This is a projection-design failure rather than a failure of the categorical theorem.

Over-strict admissibility. If the verifier rejects nearly all non-identity traces, the category degenerates toward identities and the metric balls grow slowly or not at all.

Loose admissibility. If defect is large enough that nearly all proposals are accepted, the category approaches a free trace category and the metric loses governing force.

11 Conclusion

The Coh category is a category of earned morphisms: traces become arrows only after satisfying verifier acceptance and the governing resource inequality. Projection from hidden realizations induces a semantic envelope defect, and this envelope is subadditive under observable composition. Certified endpoint-typed morphisms compose into a category with locally preordered hom-sets. Taking the infimum of envelope defect over admissible traces yields a directed extended metric. Thus Coh supports a coherent structure linking verification, category theory, projection-induced cost, and admissible geometry.

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